Script to add a node license

ONTAP Select

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Script to add a node license

You can use the following script to add a license for an ONTAP Select node.

```
#!/usr/bin/env python
##-----------------------------------------------------------------------
#
# File: add_license.py
#
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#
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#
##-----------------------------------------------------------------------

import argparse
import logging
import json

from deploy_requests import DeployRequests


def post_new_license(deploy, license_filename):
    log_info('Posting a new license: {}'.format(license_filename))

    # Stream the file as multipart/form-data
    deploy.post('/licensing/licenses', data={},
                files={'license_file': open(license_filename, 'rb')})

    # Alternative if the NLF license data is converted to a string.
    # with open(license_filename, 'rb') as f:
    #    nlf_data = f.read()
    #    r = deploy.post('/licensing/licenses', data={},
    #                    files={'license_file': (license_filename,
```

```
    nlf_data)})
 39
 40
 41 def put_license(deploy, serial_number, data, files):
 42     log_info('Adding license for serial number: {}'.format
(serial_number))
 43
 44     deploy.put('/licensing/licenses/{}'.format(serial_number), data
=data, files=files)
 45
 46
 47 def put_used_license(deploy, serial_number, license_filename,
ontap_username, ontap_password):
 48     ''' If the license is used by an 'online' cluster, a
username/password must be given. '''
 49
 50     data = {'ontap_username': ontap_username, 'ontap_password':
ontap_password}
 51     files = {'license_file': open(license_filename, 'rb')}
 52
 53     put_license(deploy, serial_number, data, files)
 54
 55
 56 def put_free_license(deploy, serial_number, license_filename):
 57     data = {}
 58     files = {'license_file': open(license_filename, 'rb')}
 59
 60     put_license(deploy, serial_number, data, files)
 61
 62
 63 def get_serial_number_from_license(license_filename):
 64     ''' Read the NLF file to extract the serial number '''
 65     with open(license_filename) as f:
 66         data = json.load(f)
 67
 68         statusResp = data.get('statusResp', {})
 69         serialNumber = statusResp.get('serialNumber')
 70         if not serialNumber:
 71             log_and_exit("The license file seems to be missing the
serialNumber")
 72
 73         return serialNumber
 74
 75
 76 def log_info(msg):
 77     logging.getLogger('deploy').info(msg)
```

```


def log_and_exit(msg):
    logging.getLogger('deploy').error(msg)
    exit(1)


def configure_logging():
    FORMAT = '%(asctime)-15s:%(levelname)s:%(name)s: %(message)s'
    logging.basicConfig(level=logging.INFO, format=FORMAT)
    logging.getLogger('requests.packages.urllib3.connectionpool').setLevel(logging.WARNING)


def main(args):
    configure_logging()
    serial_number = get_serial_number_from_license(args.license)

    deploy = DeployRequests(args.deploy, args.password)

    # First check if there is already a license resource for this serial-number
    if deploy.find_resource('/licensing/licenses', 'id', serial_number):

        # If the license already exists in the Deploy server, determine if its used
        if deploy.find_resource('/clusters', 'nodes.serial_number', serial_number):

            # In this case, requires ONTAP creds to push the license to the node
            if args.ontap_username and args.ontap_password:
                put_used_license(deploy, serial_number, args.license,
                                 args.ontap_username, args.ontap_password)
            else:
                print "ERROR: The serial number for this license is in use. Please provide ONTAP credentials."
        else:
            # License exists, but its not used
            put_free_license(deploy, serial_number, args.license)
    else:
        # No license exists, so register a new one as an available license for later use
        post_new_license(deploy, args.license)
```

```python


def parseArgs():
    parser = argparse.ArgumentParser(description='Uses the ONTAP Select Deploy API to add or update a new or used NLF license file.')
    parser.add_argument('-d', '--deploy', required=True, type=str, help='Hostname or IP address of ONTAP Select Deploy')
    parser.add_argument('-p', '--password', required=True, type=str, help='Admin password of Deploy server')
    parser.add_argument('-l', '--license', required=True, type=str, help='Filename of the NLF license data')
    parser.add_argument('-u', '--ontap_username', type=str,
                        help='ONTAP Select username with privelege to add the license. Only provide if the license is used by a Node.')
    parser.add_argument('-o', '--ontap_password', type=str,
                        help='ONTAP Select password for the ontap_username. Required only if ontap_username is given.')
    return parser.parse_args()

if __name__ == '__main__':
    args = parseArgs()
    main(args)
```

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